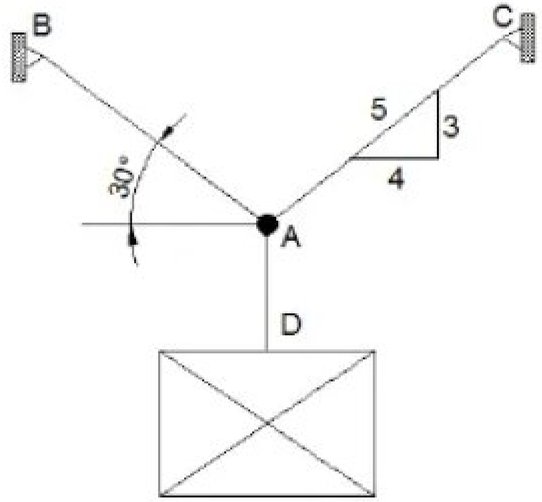
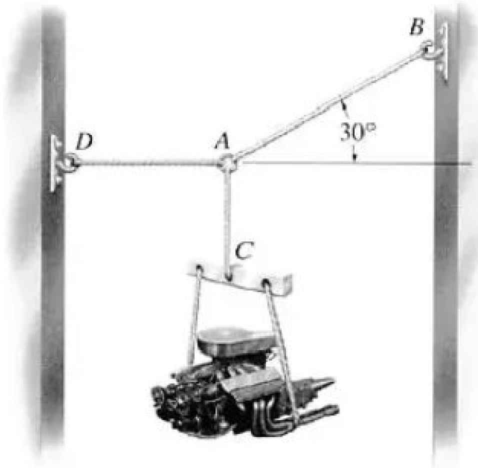
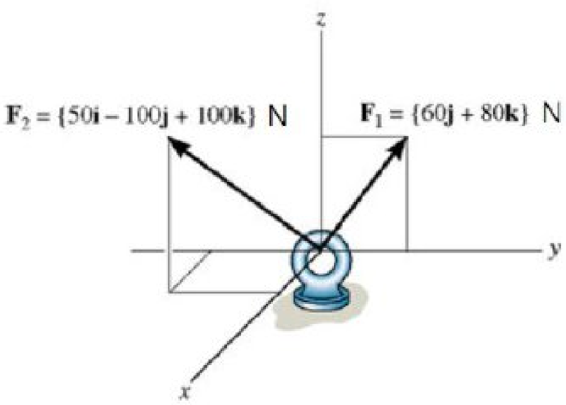

B
C
5
3
4
30°
A
D


B
D
A
30°
C


z
$\mathbf{F}_2 = \{50\mathbf{i} - 100\mathbf{j} + 100\mathbf{k}\}$ N
$\mathbf{F}_1 = \{60\mathbf{j} + 80\mathbf{k}\}$ N
y
x

20) Dados os pontos A(-1, 2), B(3, - 1) e C(-2, 4), determinar o ponto D de modo que **CD** = 0,5 **AB**. Observação: considere as letras em negritos como vetores.

21) Dados os vetores $\vec{u} = (2, -4)$, $\vec{v} = (-5, 1)$ e $\vec{w} = (-12, 6)$, determinar a_1 e a_2 tais que $\vec{w} = a_1\vec{u} + a_2\vec{v}$.

22) Calcular os valores de a para que o vetor $\vec{u} = (a, -2)$ tenha módulo 4.

23) Calcular o ângulo entre os vetores $\vec{u}=(1,1,4)$ e $\vec{v}=(-1,2,2)$.

t ff ff x x ff t t t
ff x x t o ff ff
x t ff

x ff ff ff t ff

x ff t ff x t

x ff ff ff ff ff ff t ff

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t o ﬀ ﬀ x ﬀ ﬀ ﬀ

t ﬀ t x x x x ﬀ ﬀ t s
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x ﬀ ꞓk ﬀ x t
t

x t ﬀ t x t ꞓk x
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t t x ff ȼh t ff

t ff t t x ts
x ff ff t ȼk